THE RELATIONSHIP OF DESIGN THINKING WITH 21ST CENTURY LEARNING SKILLS AMONGST STUDENTS OF ENGINEERING PROGRAMS IN MALAYSIA POLYTECHNICS

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ABSTRACT

The purpose of this research is to identify the current level of 21st century learning skills and level of Design Thinking among the polytechnic students around Malaysia. Another purpose of this research is to identify the relationship of Design Thinking with 21st century learning skills. Total of 335 students of polytechnic around Malaysia are willing to participate in this research by answering the questionnaire made for this research, which the students have been selected with simple random sampling through online method such as official email and Facebook, since face to face approach is not possible to be done during pandemic. The data has been collected for further analysis such as using Statistical Packages for Social Sciences (SPSS) software. The results show that majority of the polytechnic students are having high level of communication skills, high level of creative thinking, medium level of critical thinking, high level of collaboration skills, and high level of character development, in perspective of level of 21st century learning skills. Majority of polytechnic students also have high level of empathic, high level of defining problem, high level of ideation, high level of prototyping, and high level of product testing, in perspective of level of Design Thinking. Inferential analysis shows that there is no significant different of 21st century learning skills level and Design Thinking level among gender, courses, and year of study. In addition, there is a significant relationship of 21st century learning skills with Design Thinking, with strong correlation. Next, further discussion is done on implication of the study and recommendation of future research regarding Design Thinking in TVET institutions.

Keywords: 21st Century Learning Skills, Design Thinking, TVET, Polytechnic

INTRODUCTION

Malaysia's Ministry of Education has been stressing lots on 21st century learning among the teachers and students (Kementerian Pendidikan Malaysia, 2019). Since it is being introduced during Malaysia's Education Development Plan 2013-2025 (Mohd Rusdin and Ali, 2019), 21st century learning basically implementing the recent ways of learning and teaching in classroom especially in teaching aids and environment, in advance can increase the quality of the students on par with global benchmark, especially fulfilling industrial or society needs. Malaysia's Ministry of Education has a good attention of all the education fields, including TVET field on issues of graduates' quality. That is why 21st century learning takes part as an alternative for better education. Stated by Ministry of Educations in Malaysia's Education Development Plan (Ministry of Education of Malaysia, 2013), six main characteristics needed by students in ensuring good benchmark for global are Knowledgeable, Thinking Skills, Leadership Skills, Bilanguage Skills, Ethics and Moral Values, and National Identity.

21st learning can be executed based on few characteristics, which are student centred learning, computer as teaching aid, active learning among students, self-learning, conducive learning environment, students that understand and obey the instructions from teacher, respect culture among students and teachers, students that are responsible on their learning, achievement-based assessment, and collaborative learning (Apakah Pendidikan Abad Ke-21, 2017).

Based on the 21st century learning, few aspects are being targeted to be applied by the students at the end of the day. Teachers that apply 21st century learning should be able to create students that can communicate well, have good collaborative skills, critical thinking, creative thinking, and moral values and ethics practice among students (Apa Itu Pembelajaran Abad Ke 21 (PAK21), 2018). All these five elements are important in ensuring the best ready-to work graduates for industry or society. The real question is, do TVET learning system has the best characteristics to fulfil the elements of the 21st century learning? Do TVET graduates has the employability skills that has been decided by the industries?

Design Thinking is a process of problem solving analytically and creatively through obtaining data by experiments for creating a prototype and testing, followed by feedbacks from targets of the prototype test field (Razzouk and Shute, 2012). Although Design Thinking is claimed to be no standardized model (Waidelich et al., 2018), but we will be focusing on Design Thinking Model proposed by Hasso Plattner Institute of Design, because the model has the steps that commonly mentioned on others model, which are Ideate and Prototype (Waidelich et al., 2018). This model basically has 5 stages, proposed by Hasso Plattner Institute of Design (Friis Dam and Siang Teo, 2020), which are Empathise, Define, Ideate, Prototype and Test. Each one of them is done for the sake of problem solving and product development. Design Thinking usually being used in industry for Research and Development purpose.

Referring to the brief explanation on Design Thinking module, it is no doubt that this module may have association with the nature of learning in TVET institutions, where certain assessments done by the students are project-based learning. This is good news for TVET institutions since the module should have high potential in producing TVET graduates that are not only capable on psychomotor intelligence, but also on problem solving intelligence.

MAIN RESULTS

This research mainly done to find a few objectives, which are to find the level of 21st century learning skills, the level of Design Thinking, and the relationship of 21st century learning skills with Design Thinking. In this chapter, we will be discussing on the results obtained for polytechnics with students that take Diploma in Engineering from any field. There will be descriptive analysis and inferential analysis done for answering the research questions. Items related in the instrument consist of characteristics for both 21st century learning skills and Design Thinking, which the score will be analysed in SPSS software.

Level of 21st Century Learning Skills and Design Thinking amongst Engineering Polytechnic Students

The level of 21st Century Learning Skills and Design Thinking are distinguished into three level, which are Low level, Medium Level, and High Level. Below are the details on how the level of mastery is distinguished:

Table 1. Score range of mastery level

Mastery Level	Score Range
Low level	1 – 2.33
Medium level	2.33 – 3.67
High level	3.67 - 5

SPSS software has been used to help the data analysis of total 335 polytechnic students that are willing to participate in this research, which it can be concluded that the samples size can approximately represent 2600 polytechnic students' population (Krejcie and Morgan, 1970).

Sampling method used is simple random sampling, which the best way of Likert Scale questionnaire distribution is through Facebook contacts and group that have any relations with polytechnic, such as student representative council of the polytechnic Facebook Page, since it is difficult to obtain data from all available polytechnics or polytechnic of interest, and not all polytechnics are able to cooperate with the research due to the pandemic situation where social distancing is the top priorities at the moment (Soalan Lazim (FAQ) Berkaitan Perintah Kawalan Pergerakan Bersyarat (PKPB) Kementerian Pengajian Tinggi, 2020).

Table 2. Distribution of respondents on respective level of 21st century learning skills dimension.

Skills Dimension	Frequency			Percentage		
	Low	Medium	High	Low	Medium	High
Communication	17	129	189	5.1	38.5	56.4
Creative Thinking	17	95	223	5.1	28.4	66.6
Critical Thinking	23	178	134	6.9	53.1	40.0
Collaboration	13	67	255	3.9	20.0	76.1
Character Development	16	79	240	4.8	23.6	71.6

From all the analysis in Table 2, it is identified on all highest percentage of mastery level of all the 21st century learning skills, where majority of the students have high level of communication skills, high level of creative thinking skills, medium level of critical thinking skills, high level of collaboration skills, and high level of character development.

Table 3. Distribution of respondents on respective level of Design Thinking stages dimension.

Stages Dimension	Frequency			Percentage		
	Low	Medium	High	Low	Medium	High
Empathise	13	69	253	3.9	20.6	75.5
Define	14	76	245	4.2	22.7	73.1
Ideation	15	101	219	4.5	30.1	65.4
Prototype	12	122	201	3.6	36.4	60.0
Test	15	89	231	4.5	26.6	69.0

From all the analysis, it is identified on all highest percentage of mastery level of all the Design Thinking characteristics, where majority of the students have high level of empathic, high level of defining problem, high level of ideation, high level of prototyping, and high level of product testing.

The Relationship of Design Thinking With 21st Century Learning

Pearson Correlation test is needed to identify the relationship between 21st century learning skills and Design Thinking. It is mentioned before that Design Thinking process may have a necessary skills or characteristics needed to fulfil it, thus seeing 21st century learning skills as the requirement to execute Design Thinking. Thus, Pearson Correlation can clear the argument on how both related. First, it is needed to identify the type of data first. If the data is ordinal scale, usually non-parametric is used such as Spearman Correlation. Referring to argument on Likert scale from Carifio and Perla (2007), Likert scale can be treated as interval scale data (Carifio & Perla, 2007). Thus, Pearson Correlation is proceeded. Then, it is possible to do Pearson correlation. To run this analysis, hypothesis is made as follow:

Table 4. Hypothesis for Pearson Correlation test

H_0	There is no correlation on Design Thinking with 21st century learning skills.
H_1	There is a correlation on Design Thinking with 21st century learning skills.

The correlation analysis is done on 21st century learning skill score with Design Thinking score that associated with related skills resembling 21st century learning skill. Here are the results from the Pearson correlation test:

Table 5. Pearson Correlation of Communication and Communication in Design Thinking

Correlations

		Communication Score	Communication in Design Thinking Score
Communication Score	Pearson Correlation	1	.772**
	Sig. (2-tailed)		.000
	N	335	335
Communication in Design Thinking Score	Pearson Correlation	.772**	1
	Sig. (2-tailed)	.000	
	N	335	335

**. Correlation is significant at the 0.01 level (2-tailed).

The results for the correlation test show that Pearson Correlation is 0.772 with a significance two tailed value of 0.0001. It means that there is a very strong correlation ($r = 0.772$), and has a significant correlation, as $p < 0.05$, for the communication score and communication in Design Thinking score. Hence, we must reject the hypothesis null.

Table 6. Pearson Correlation of Creative and Creative in Design Thinking

Correlations

		Creative Score	Creative in Design Thinking Score
Creative Score	Pearson Correlation	1	.757**
	Sig. (2-tailed)		.000
	N	335	335
Creative in Design Thinking Score	Pearson Correlation	.757**	1
	Sig. (2-tailed)	.000	
	N	335	335

**. Correlation is significant at the 0.01 level (2-tailed).

The results for the correlation test show that Pearson Correlation is 0.757 with a significance two tailed value of 0.0001. It means that there is a very strong correlation ($r = 0.757$), and has a significant correlation, as $p < 0.05$, for the creative thinking score and creative in Design Thinking score. Hence, we must reject the hypothesis null.

Table 7. Pearson Correlation of Critical and Critical in Design Thinking

Correlations

		Critical Score	Critical in Design Thinking Score
Critical Score	Pearson Correlation	1	.740**
	Sig. (2-tailed)		.000
	N	335	335
Critical in Design Thinking Score	Pearson Correlation	.740**	1
	Sig. (2-tailed)	.000	
	N	335	335

**. Correlation is significant at the 0.01 level (2-tailed).

The results for the correlation test show that Pearson Correlation is 0.740 with a significance two tailed value of 0.0001. It means that there is a very strong correlation ($r = 0.740$), and has a significant correlation, as $p < 0.05$, for the critical thinking score and critical in Design Thinking score. Hence, we must reject the hypothesis null.

Table 8. Pearson Correlation of Collaboration and Collaboration in Design Thinking

Correlations

		Collaboration Score	Collaboration in Design Thinking Score
Collaboration Score	Pearson Correlation	1	.817**
	Sig. (2-tailed)		.000
	N	335	335
Collaboration in Design Thinking Score	Pearson Correlation	.817**	1
	Sig. (2-tailed)	.000	
	N	335	335

**. Correlation is significant at the 0.01 level (2-tailed).

The results for the correlation test show that Pearson Correlation is 0.817 with a significance two tailed value of 0.0001. It means that there is a very strong correlation ($r = 0.817$), and has a significant correlation, as $p < 0.05$, for the collaboration score and collaboration in Design Thinking score. Hence, we must reject the hypothesis null.

Table 9. Pearson Correlation of Character and Character in Design Thinking

Correlations

		Character Score	Character in Design Thinking Score
Character Score	Pearson Correlation	1	.722**
	Sig. (2-tailed)		.000
	N	335	335
Character in Design Thinking Score	Pearson Correlation	.722**	1
	Sig. (2-tailed)	.000	
	N	335	335

**. Correlation is significant at the 0.01 level (2-tailed).

The results for the correlation test show that Pearson Correlation is 0.722 with a significance two tailed value of 0.0001. It means that there is a very strong correlation ($r = 0.772$), and has a significant correlation, as $p < 0.05$, for the character development score and character in Design Thinking score. Hence, we must reject the hypothesis null.

Based on the results obtained from Pearson Correlation test, it tells that each variable of 21st century learning skills has a strong correlation with each variable of associated skills that are implemented in Design Thinking and has a significant relationship.

CONCLUSION

Based on the research findings, polytechnic students have showed a good significant result on their mastery level on 21st century learning, except for critical thinking skill, which is in medium mastery level. Polytechnic should have taken note on the critical thinking development of the students since there are findings that show the importance of having good critical thinking in employability (Maknun and Siahaan, 2018; Muharram et al., 2019). Essentially, research question one has been answered with finding of high mastery level in 21st century learning skills among the polytechnic students except on critical thinking.

Polytechnic students also showed a good significant result on their mastery level on characteristics of designer by Owen (2007), and Razzouk and Shute (2012). As mentioned before, this is a good sign for the future of Design Thinking implementation in TVET institutions, where students are impliedly possessing a set skills or characteristics of a designer. There are studies that show the benefits of Design Thinking in the industry (Hrovatin et al., 2008; Ruchira Prasad et al., 2018; Taratukhin et al., 2020) and in education (Deitte and Omary, 2019; Henriksen et al., 2020; Sándorová et al., 2020). Thus, it can be said that research question two is answered with the results of high master level in characteristics of Design Thinking among the polytechnic students.

The main results on finding the relationship of Design Thinking with 21st century learning skills show a very good signification correlation between those variables. It can be concluded that if students have a high possession of characteristics of designer, the students can have high level of 21st century learning skills. Research question three has been answered with strong correlation of Design Thinking and 21st century learning skills. For the future of TVET, it is suggested for highest TVET management to support the future research on Design Thinking implementation in classroom of TVET. New curriculum arrangement or improvement of syllabus by adding the generated module of learning, originated by Design Thinking Process can become reality in the future for better TVET graduates.

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